Vishak K Bhat

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Education

IIT(ISM) Dhanbad May 2026
Integrated Master of Technology in Mathematics and Computing (GPA: 8.85 / 10.00, 3rd Rank) *Dhanbad, Jharkhand*

- **Relevant Coursework:** Data Structures and Algorithms (C++), Object Oriented Programming , Data Base Management System , Data-Mining , Prob & Stats , Linear Algebra , Differential equations, Computational Finance, Optimization Techniques, Numerical Methods, Numerical Optimization

Publications

interwhen: A Generalizable Framework for Verifiable Reasoning with Test-time Monitors arXiv
Vishak K Bhat, Subbarao Kambhampati, Vineeth N. Balasubramanian, Nagarajan Natarajan, Amit Sharma
NeurIPS 2026 Under Review
ICLR 2026 Workshop VerifAI-2
ICLR 2026 Workshop LLM Reasoning

Genesis: Towards Explainable Causal Discovery
Vishak K Bhat, Abhinav Thorat, Ravi Kumar Kolla, Niranjan Pedanekar
AAAI 2027 Under Review

GNN For Muon Particle Momentum estimation arXiv [GSoC'24]
Vishak K Bhat, Eric A. F. Reinhardt, Sergei Gleyzer
arXiv preprint

Experience

Microsoft Research – Research Intern → Research Fellow (Mentor: Dr. Amit Sharma) November - present

- Designed and open-sourced **interwhen**, a test-time scaling method that **verifies intermediate reasoning steps** and **steers the model** toward the correct answer rather than checking only the final answer.
- Introduced **asynchronous fork-and-verify reasoning**, where verification runs in parallel with the main LLM stream, adding negligible latency while keeping total token usage, including verification, comparable to CoT.
- Achieved state-of-the-art performance over Tree of Thought, Best of N, Generate-Test, and Chain of Thought.
- Demonstrated gains on agentic (Tau2Bench, Agent Safety Bench, VITA Bench) and non-agentic (Maze, Spatial Map, ZebraLogic, VERINA) benchmarks across 3B to frontier LLMs.
- Another use case of **interwhen**: **early stopping**, which halts the reasoning process when the model is sufficiently confident, reducing token usage and inference cost.

Sony Research India – Research Intern June –November

- Improving upon **CausalOrder** and **LLM_CD** by developing a novel LLM-driven causal discovery framework.
- Enhancing causal discovery through parallel use of **Partial Ancestral Graphs (PAG)** with LLM-guided search, benchmarked on standard datasets including Asia, Cancer, Earthquake, Child, and Survey.

Swiggy – Data Scientist Intern Aug – May 2025

- Optimized Surge Values: Solved the **Multi-Armed Bandit (MAB)** problem by training a model using Policy Grad increasing the show up rate by 5% with no increase in expense.
- Designed an algorithm to rezone the existing zones to minimize cross-zone orders by 6%.

Google Summer of Code – ML4SCI - CERN (GSoC'24) April – Sept 2024

- Implemented two GNN architectures that outperformed transformer models in predicting the momentum of Muon particles. One model improved inference speed by 27%, while the other reduced mean squared error by 14%.
- Performed 3 methods to convert the data into graphs and then created a custom Message Passing Layer

NyunAI –Research Intern May – June 2024

- Developed Nyun-Llama3-62B, a model with 12% fewer parameters than Llama-3-70B, achieving comparable or superior performance across MMLU, Winogrande, BoolQ, Hellaswag, Arc Challenge, and GSM8K benchmarks.

Bosch Global Software Technologies – Computer Vision Research Intern Jan – May 2024

- Used Maskformer, Mask2former models to obtain Semantic Segmentation of the dataset achieving F1 score of 0.89
- Performed object detection using groundingDINO to get the masks of objects. Utilized Stable Diffusion inpainting to modify real world obstacle images, generating a diverse dataset of 10,000 synthetic images for model fine-tuning.

UiT, The Arctic University of Norway - Bio-AI Labs – Research Intern Oct – Dec 2023

- Designed a model to process variable-size images by converting them into graphs, enabling further analysis for benchmarking tasks. Achieved an F1 score of 0.92 on the Bengali text dataset by graph classification

Projects

Vesuvius Challenge – Ink Detection (Kaggle Bronze Medal) | *Computer Vision, Image Processing, PyTorch* | GitHub

- Segmented ink regions in carbonized papyrus fragments, achieving an F-beta score of 0.74.
- Trained multiple **U-Net** architectures using segmentation-models-pytorch with **ResNet, DenseNet, SE-ResNet, VGG, and Inception** backbones.
- Built ensemble models and applied **Test-Time Augmentation (TTA)** to improve generalization and leaderboard performance.

Question Vs Answer Agents | *Multi-Agent AI, LLM, Reinforcement Learning, Python* | GitHub

- Built competing Q-agent and A-agent using LLMs for a head-to-head AI tournament; secured **1st place** among competing agent teams.
- Improved performance via supervised fine-tuning, GRPO-based reinforcement learning, and self-play strategies.
- Curated synthetic datasets and engineered prompts to enhance reasoning accuracy and structured output compliance.

Personally Identifiable Information (PII) classification | *NLP, Attention, HuggingFace, Transformers, Finetuning*

- Finetuned BERT to identify PII by masked LM, then finetuned it to classify the type of PII achieving F1 score of 0.92

Vision Transformers | *NLP, Attention, Computer Vision, Pytorch, Azure, HuggingFace* project-link

- Trained Vision Transformer (ViT) model to distinguish between electrons and photons, achieving a ROC AUC of 0.94
- Used **Pre-processing** methods to improve the quality of the matrix data increasing the accuracy by 8%
- Used **Azure, Google Cloud Virtual Machines** to train these huge models

Technical Skills

Languages: C++, C, Python **Technologies**: Pytorch, Matplotlib, OpenCV, Scikit-Learn, Transformers

Concepts: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, LLM reasoning, Diffusion models, Reinforement Learning, Multimodals, VLMs, Data Structures and Algorithms, Time series, Model pruning, Causal Discovery, Finetuning, Distillation

Achievements and Awards

- Kaggle Competition Expert – Top 0.1% (185K+ users), 3× Bronze Medalist
- Won 1st Prize worth **Rs. 1.5 Lakhs** at the AMD-organized AI Premier League Hackathon held at IISc Bangalore
- Attended the ICLR'25 in Rio De Janerio, Brazil
- Winner, AgenticAI Hackathon: Secured 1'st place in a Microsoft-organized hackathon
- 1st Place, Winter of Code (WOC): Ranked 1st out of 1000+ participants in a highly competitive hackathon.
- Scored in the top 0.03% in JEE Advanced 2021 and secured 240th rank in the Karnataka CET exam 2021.
- Selected to present my work at the GSoC'24 Summarization Program among all GSoC students.
- Lead, ML Division – CyberLabs: Mentored juniors through teaching sessions and earning high recognition.